

AREarthQuakeDrill: Toward Increased Awareness of Personnel during Earthquakes via AR Evacuation Drills

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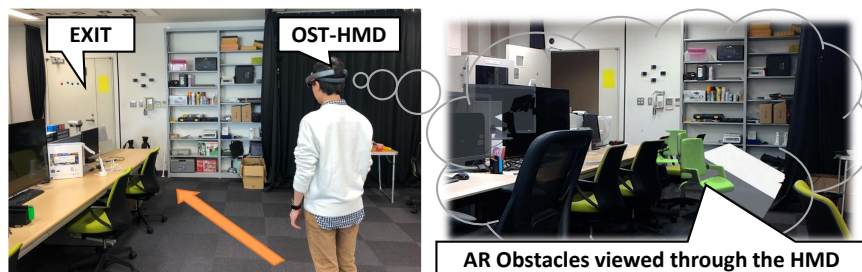


Figure 1: Our proposed evacuation drill shows virtual obstacles in Augmented Reality. We use an OST-HMD to display the aftermath of an earthquake along with the real scene. As a result, users can perform evacuation drills safely and be fully aware of the surrounding environment and obstacles.

ABSTRACT

Evacuation drills are carried out to reduce the injury or death caused by earthquakes. However, the content of evacuation drills is generally predetermined for specific evacuation routes and actions. This inflexibility can reduce user motivation and sincerity.

In this paper, we propose an Augmented Reality (AR) based evacuation drill system. We use an optical see-through head-mounted display (HMD) for mapping a room and recognizing interior. Our system constructs an AR drill environment of the real environment, and reproduces the after-effects of an earthquake by applying vibrations to the objects. We evaluated our system in an experiment with 10 participants. Comparing cases with and without AR obstacles, we found that AR training affected participant motivation and the diversity of traversed evacuation routes during practice.

Index Terms: H.5.2 [User Interfaces]: User Interfaces—Graphical user interfaces (GUI); H.5.m [Information Interfaces and Presentation]: Miscellaneous

1 INTRODUCTION

In many countries around the world, large-scale earthquakes have caused severe damage and injury to inhabitants. Though we often focus on damages such as collapsed buildings, another large problem is that people have trouble evacuating due to damage caused by fallen or broken furniture, equipment, or indoor items [4].

Annual evacuation drills have been created to improve well-rehearsed knowledge of procedures for evacuation. However, these evacuation drills have several inherent problems. First, it does not reproduce unexpected situations like those due to obstacles. Second, evacuation drills have a predefined route and pattern, so users get used to the same evacuation drill. As a result, participants may lack motivation for performing evacuation drills.

To date, researchers have proposed a number of different systems for improving disaster response. Li et al. [2] use Virtual Reality (VR)

for developing an evacuation drill application about protecting one's body in the room. Stigal et al. [3] used a mobile AR application approach that shows an evacuation route and reproduces the disaster situation on a map. However VR-based systems still have many challenges, such as a reduction in attention to surroundings and sickness [1]. Additionally, AR-based mobile applications cannot provide an immersive experience resembling disaster conditions.

To tackle these problems, in this paper we propose an evacuation training system that applies Augmented Reality (AR) to create an earthquake scenario and resulting routes for practicing evacuations. We use an Optical-see through HMD (OST-HMD) to build a training environment along with the real environment. This use of AR improves engagement of participants by more realistically replicating a disaster scenario and the accompanying tension of the situation. Our contributions are as follows:

- A simulation that can create various drill environments.
- Semi-automatic recognition of real objects and placement of corresponding virtual objects.
- Experiments showing that our system influences user escape time and route and improves user motivation for evacuation drills.

2 PROPOSED SYSTEM IMPLEMENTATION

First, object detection is implemented using Microsoft Azure's Custom Vision service, which creates a machine learning model on the cloud. The HoloLens uses images from its front-facing camera and uploads them to the cloud to get detected tags. Based on this tag, our system places a corresponding virtual object in the AR training environment.

Second, for room mapping, we use the HoloLens' spatial recognition function. An infrared-based sensor captures the surface of the room and composes a mesh-like room. By placing virtual objects in this room, the virtual training environment is overlaid and reproduced in the real world. The location of the placement is determined by the intersection of the line of sight passing through the object and the mesh.

Finally, the built environment shakes at the beginning of the evacuation drills. These vibrations cause virtual objects to fall or move, hindering the user from evacuating. By randomly setting this pseudo-earthquake, various disaster situations can be created.

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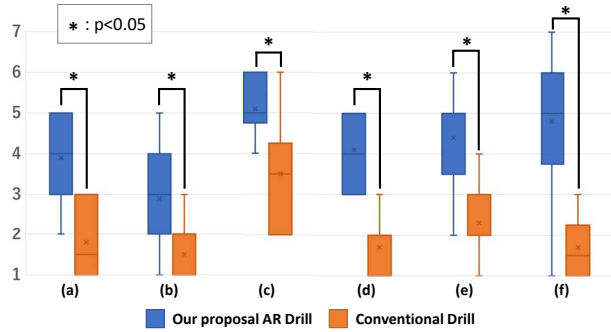


Figure 2: Subjective scoring of the six survey items.

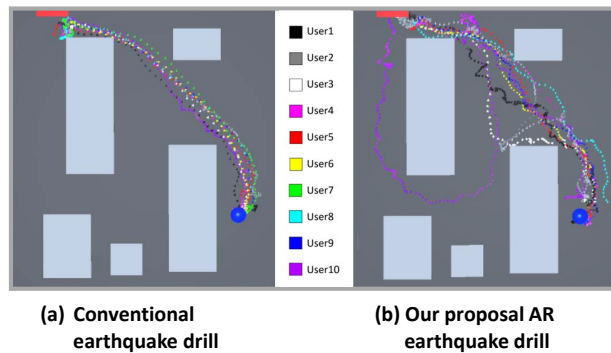


Figure 3: A plot of user exit trajectories observed during experiments without (left) and with (right) display of AR obstacles

3 EXPERIMENT

The purpose of our research was to confirm how our proposed AR evacuation drill system would affect user behavior and evacuation path. Therefore, we set up an experiment to evaluate our system using a questionnaire and quantitative data. We included 10 participants in total from a college campus (1 female, 9 males between ages 22 to 24), majoring in information science.

The experiment was set up to compare the proposed AR evacuation drill system with a conventional evacuation drill. The user worked on both drills and was instructed not to touch the AR obstacles during the proposed drill. To keep the conditions the same, the user was instructed to wear HoloLens in both drills. To alleviate ordering effects, we split the users into two groups, each starting with a different order.

After the experiment, we asked the user to complete a questionnaire about each of the two systems. There were 6 questions and they were answered using a 7-point Likert scale (with 1 being "strongly disagree" and 7 being "strongly agree").

4 RESULTS AND DISCUSSION

This section describes the analysis and results of the experimental data, followed by a summary of our contributions based on the results.

The following six questions were included on the survey:

- Did you feel nervous about the training?
- Did you feel fearful about the earthquake?
- Did you get serious about the training?
- Did you anticipate the situation at the time of disaster?
- Do you think this training is effective?
- Do you want to get the same training in the future?

We analyzed the the questionnaire responses using a Wilcoxon rank-sum test. Fig. 2 shows that T-test indicates a significant difference in the comparisons of the cases with and without AR obstacles for all questions ($p < 0.05$).

The user's evacuation paths are shown in Fig.3. The blue circle indicates the starting point and the red rectangle indicates the exit. All user paths run from the blue circle to the red rectangle, and each path colored with a unique color for each user. The light gray rectangles show the placement of real, immovable obstacles such as desks and pillars.

We obtained significant differences from the survey. That means that our proposed method has received a positive evaluation. Therefore, AR obstacles enhanced the reality of disaster reproduction in our evacuation drills. This can lead to improved user motivation and awareness, and our system can provide more effective drills than conventional approaches.

Also, in the conventional training without AR obstacles, the evacuation paths of all users are almost the same. This indicates that all users were properly trained in the traditional way and educated to follow the shortest path to the exit. On the other hand, if AR obstacles are displayed, each user's path will be different. Some people were able to get to the exit following the shortest path, while others had to make a detour. This indicates that our system contributes to provide various situations and paths in evacuation drills.

From these results, we can conclude that our proposed method affects the motivation of participants and the diversity of evacuation paths. However, some commented that the HoloLens did not work well in the dark and wearing it caused discomfort. We also received a comment that the appearance of the placed AR objects is inconsistent with real environment, which makes users feel uncomfortable.

5 CONCLUSION

In this paper, we proposed an AR evacuation drill to assist with earthquake evacuation training. By recognizing, duplicating and displaying virtual obstacles using AR, our system aims to improve user responsiveness, help increase their awareness, and provide an alternative to the typical escape patterns of a traditional evacuation drill program. Our AR evacuation drill was developed by placing virtual objects at the target object of interest in the real scene and repositioning those objects using a physics based earthquake simulation.

The results of the experiment revealed that displaying obstacles using AR had an impact on user escape trajectories and can likely improve the user's awareness during real disaster scenarios.

Future work includes providing a more realistic training environment using voice instructions, adding other implementations such as fire disasters, and reviewing the devices used.

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